

Point Patterns II

HES 505 Fall 2025: Session 16

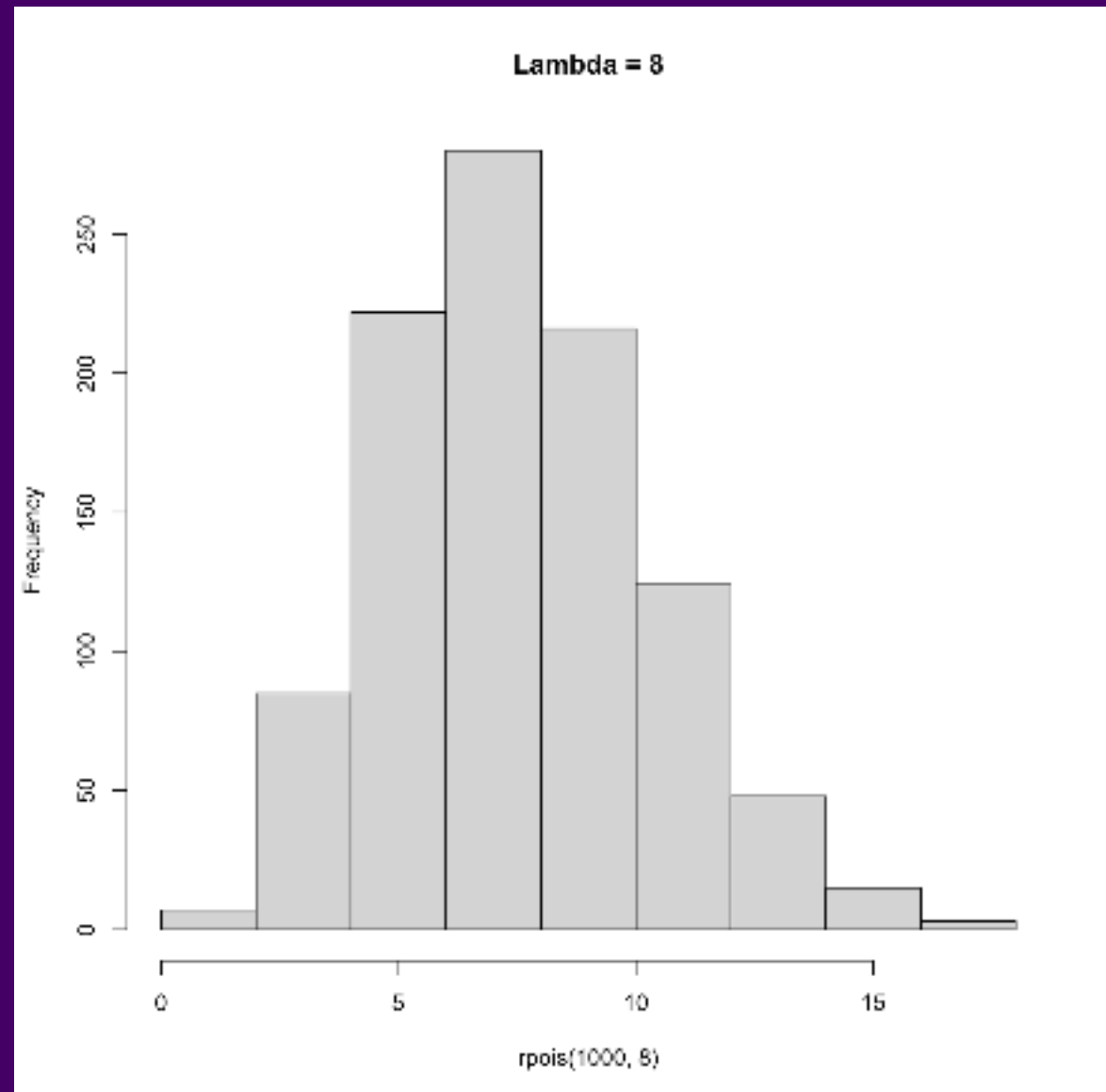
Matt Williamson

Point Processes and Why You Should Care

- They're ubiquitous
- Well-specified to many social and ecological theories
- Helps build intuition for interpolation

The Poisson Distribution

- Discrete probability distribution
- Probability of events in a fixed interval of time or space
- Mean (expectation) = variance = λ



Re-visiting CSR

What is CSR? What is it not?

What is the business case for CSR?

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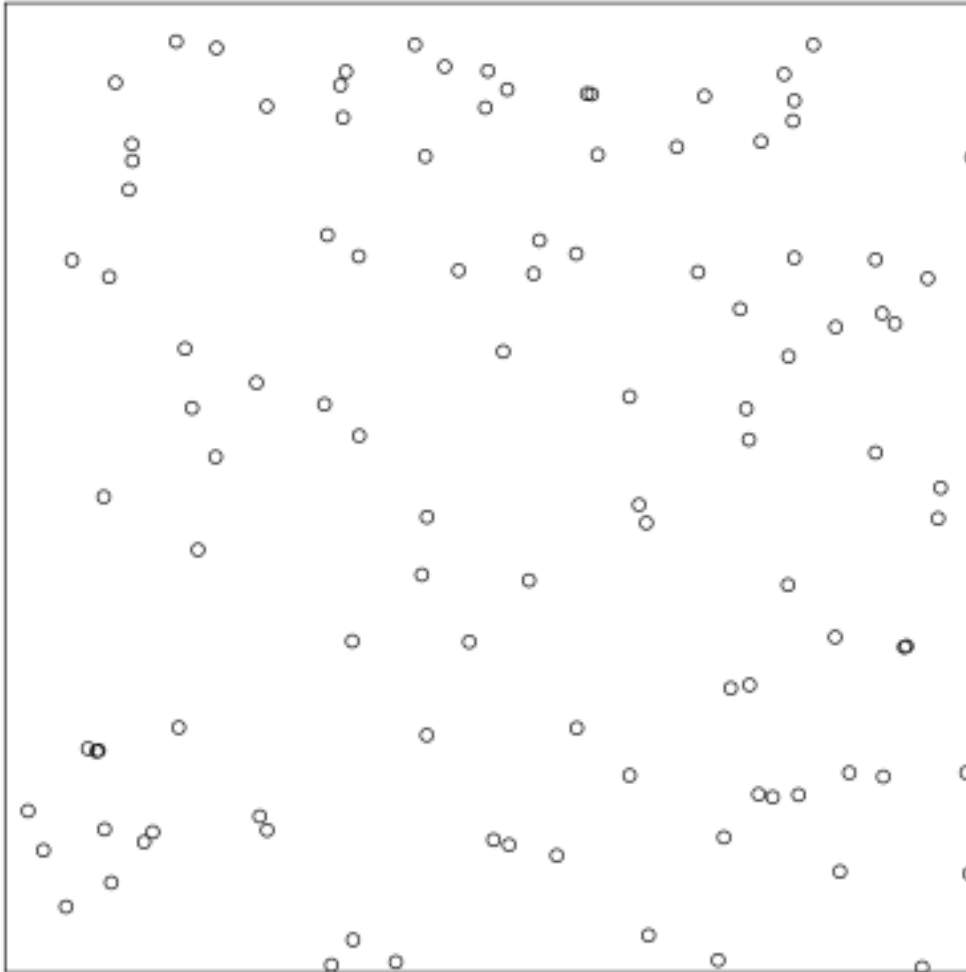
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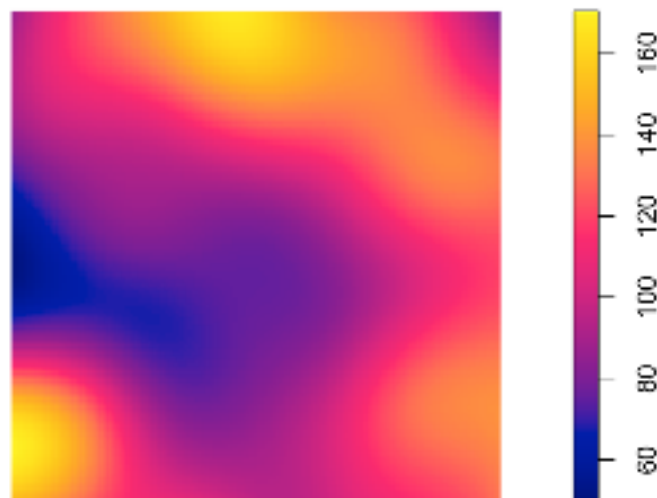
Homogeneous Poisson Process (CSR)



- Constant intensity
- Points are independent
- Density is a crude estimate of intensity

Revisting CSR

Estimated Intensity (HPP)



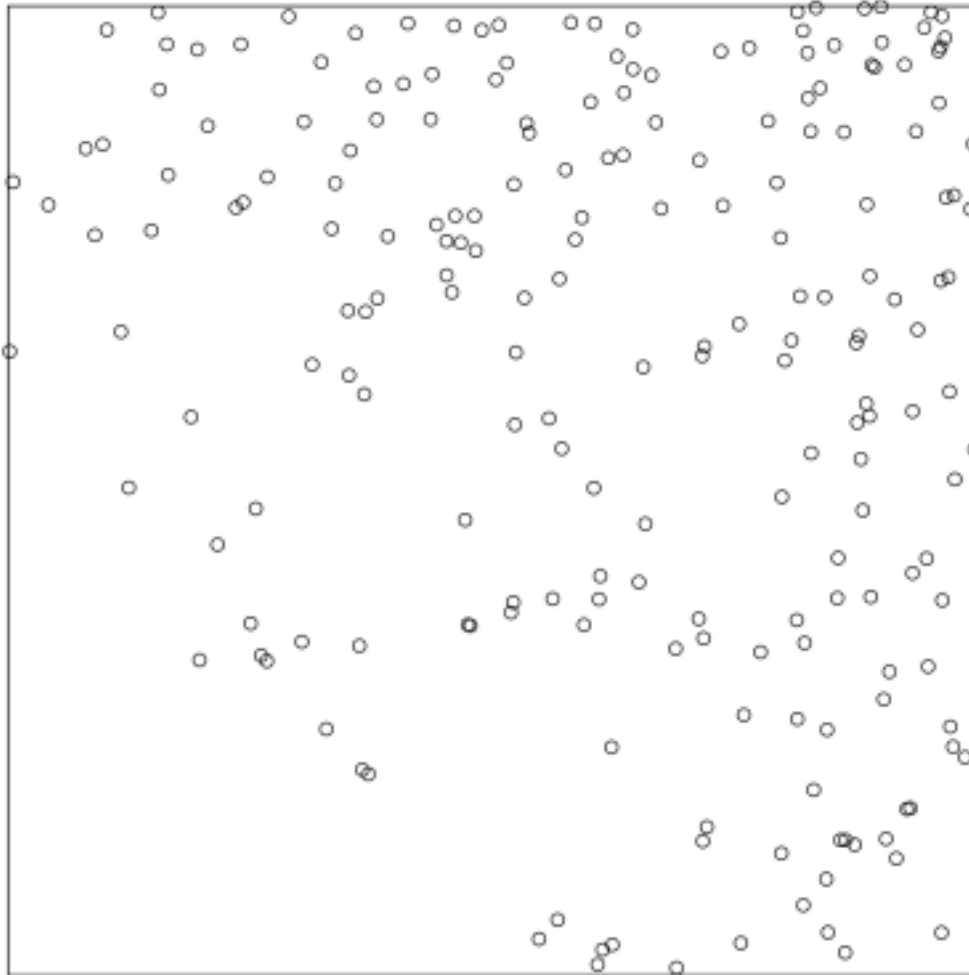
Density

0	0	100	0	0	0	0	300	200	100
200	300	100	100	100	200	0	100	300	100
100	100	0	100	100	0	0	0	200	0
200	100	200	0	200	0	100	0	0	200
300	200	100	0	0	200	100	100	0	100
100	0	300	100	0	100	0	100	300	0
200	200	0	0	100	200	0	100	0	100
100	100	100	100	200	0	0	200	300	100
200	200	200	200	100	0	200	100	200	0
0	100	100	200	0	200	200	200	100	100

First-Order Effects

- **First-order effects** are the effects of a single factor on the response variable, assuming all other factors are held constant.
- They represent the main effects of the factors being studied.
- First-order effects are typically the most important and easiest to interpret.
- They are often the focus of initial experiments and analysis.
- First-order effects can be identified using a variety of statistical methods, including:
- **ANOVA** (Analysis of Variance)
- **Regression analysis**
- **Design of experiments** (DOE)
- **Factorial designs**
- **Partial correlation**
- **Path analysis**
- **Structural equation modeling** (SEM)
- **Bayesian networks**
- **Machine learning** (e.g., linear regression, decision trees)
- First-order effects can be visualized using a variety of plots, including:
- **Bar charts**
- **Line graphs**
- **Scatter plots**
- **Interaction plots**
- **Regression lines**
- **Partial regression plots**
- **Path diagrams**
- **Bayesian networks**
- **Machine learning models** (e.g., linear regression, decision trees)

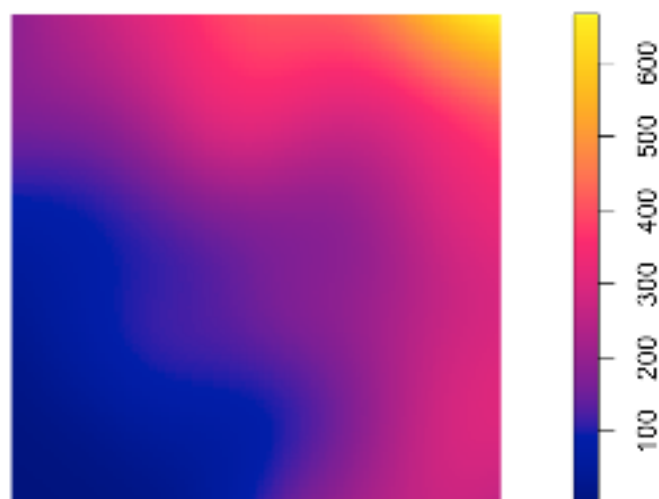
Inhomogeneous Poisson Process



- Intensity varies spatially
- Points are independent

First-Order Effects

Estimated Intensity (IPP)

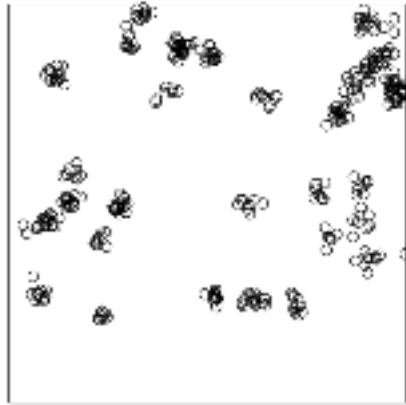


Density

0	300	200	100	0	0	0	0	0	0
500	100	100	100	200	0	100	0	0	0
200	200	200	0	0	200	300	0	0	0
300	400	200	500	100	0	200	300	0	0
500	100	800	0	0	100	200	0	0	0
400	400	300	200	300	0	400	0	0	200
700	300	100	100	100	300	200	100	0	400
200	300	200	300	0	100	300	100	300	100
1000	200	300	500	500	200	400	200	400	400
900	500	300	300	300	200	300	400	400	100

Second-Order Effects

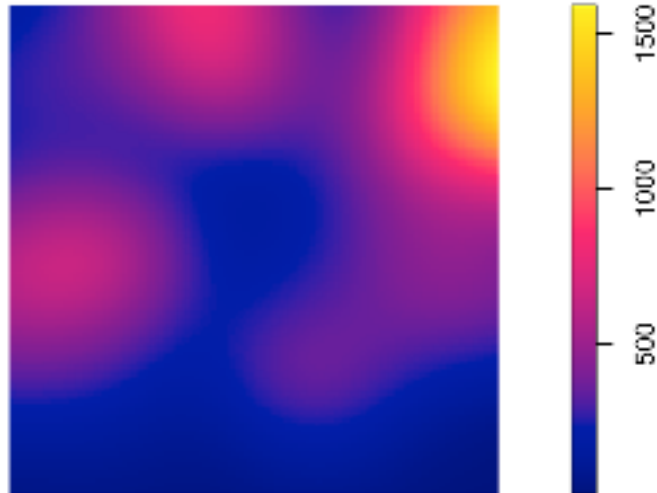
Inhomogeneous Cluster Process



- Intensity of “parents” is first-order
- Intensity of “children” is second-order

Second-Order Effects

Estimated Intensity (Clustered)



Density

0	200	0	0	0	1200	100	1000	0	0
0	1200	100	100	1700	1100	0	0	0	0
300	200	0	0	500	1900	300	800	0	0
1300	300	400	0	0	0	0	0	0	0
700	2100	300	0	0	0	0	100	0	0
0	1100	0	0	200	100	0	1500	0	0
0	0	1000	0	200	400	0	1000	0	0
0	0	0	100	900	100	100	1100	0	0
900	900	2500	0	600	1000	400	0	0	0
800	3100	2500	0	200	200	600	0	0	0

Key Points

- Poisson distribution: mean = variance = λ -> First Order
- **Homogeneous** refers to the rate, not it's outcome
- Statistical tests ask “Could this outcome be generated by a constant rate?”

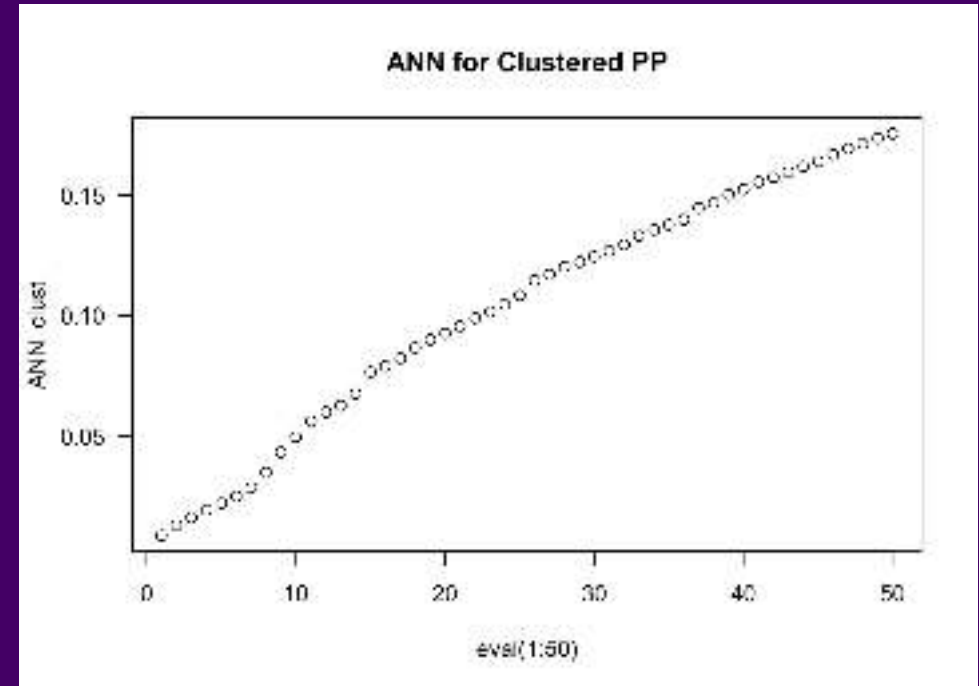
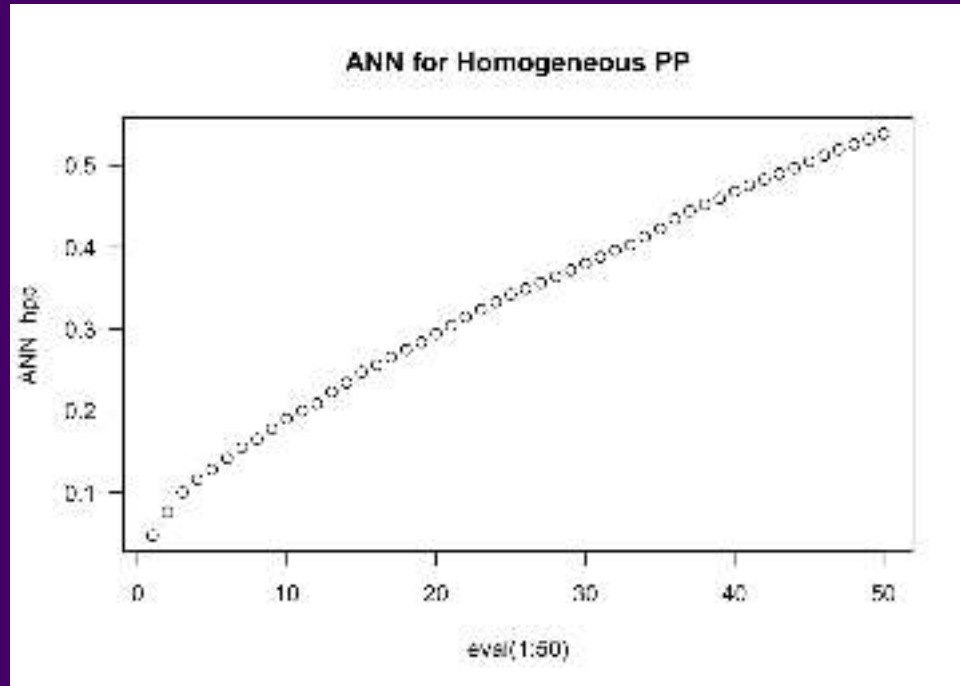
Second-Order Analysis

Distance based metrics

- Provide an estimate of the *second order* effects
- *Mean nearest-neighbor distance*:

$$\hat{d}_{\min} = \frac{\sum_{i=1}^m d_{\min}(s_i)}{n}$$

Mean Nearest Neighbor Distance



Ripley's K Function

- Nearest neighbor methods throw away a lot of information
- The K function is an alternative, based on a series of circles with increasing radius

$$K(d) = \lambda^{-1} E(N_d)$$

- We can test for clustering by comparing to the expectation:

$$K_{CSR}(d) = \pi d^2$$

- if $k(d) > K_{CSR}(d)$ then there is clustering at the scale defined by d

Ripley's K Function

- When working with a sample the distribution of K is unknown
- Estimate with

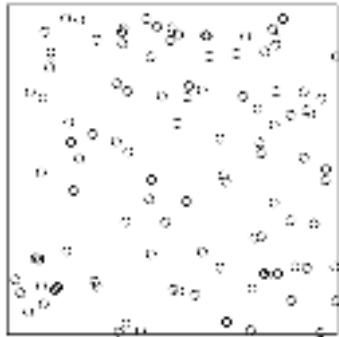
$$\hat{K}(d) = \hat{\lambda}^{-1} \sum_{i=1}^n \sum_{j=1}^n \frac{I(d_{ij} < d)}{n(n-1)}$$

where:

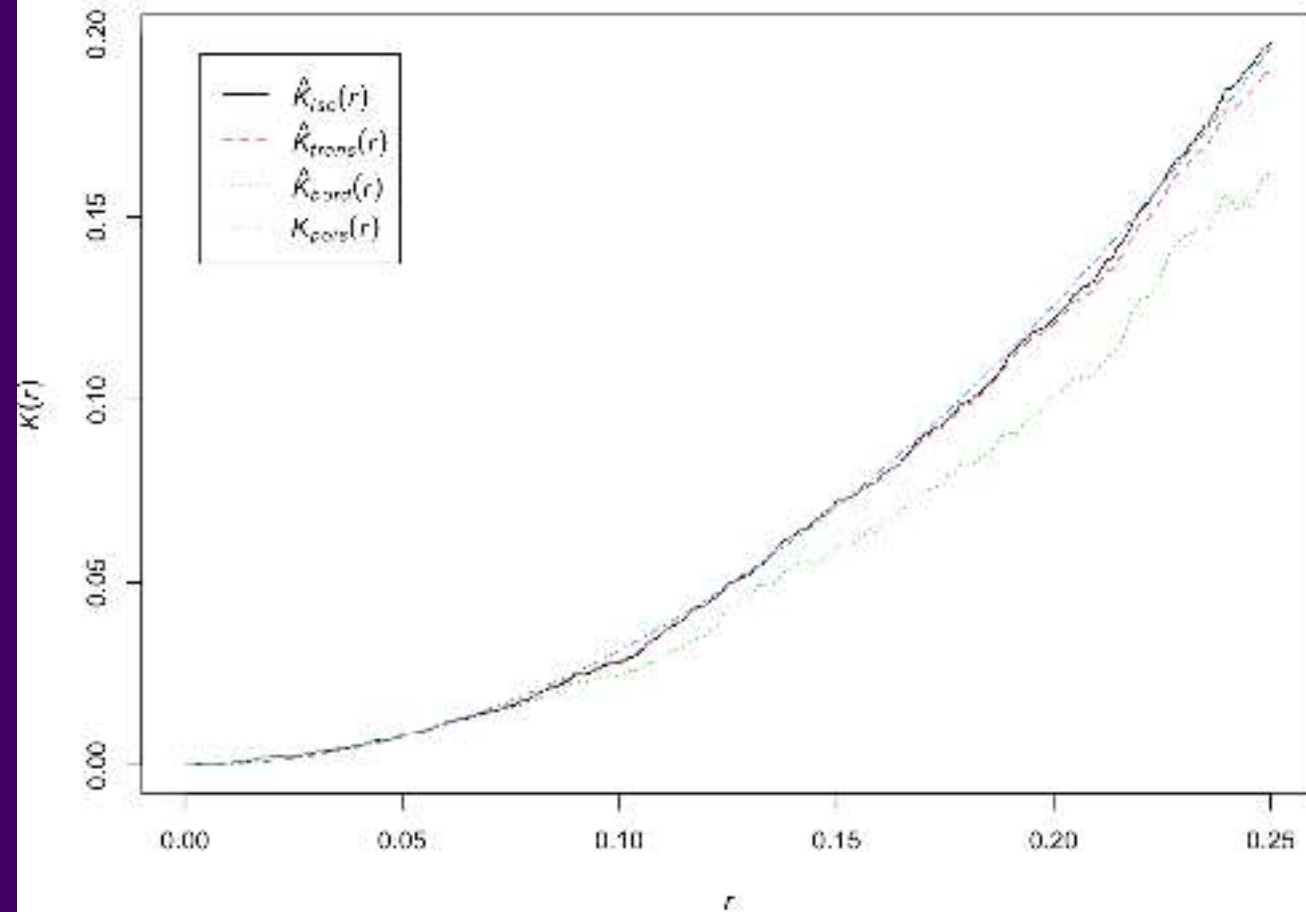
$$\hat{\lambda} = \frac{n}{|A|}$$

K for HPP

Homogeneous Poisson Process (C5R)

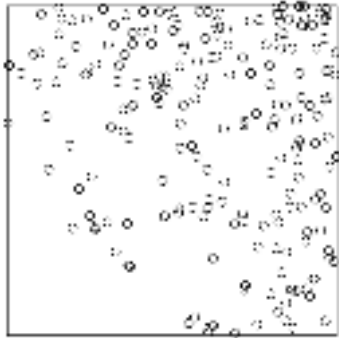


K: Homogeneous Poisson

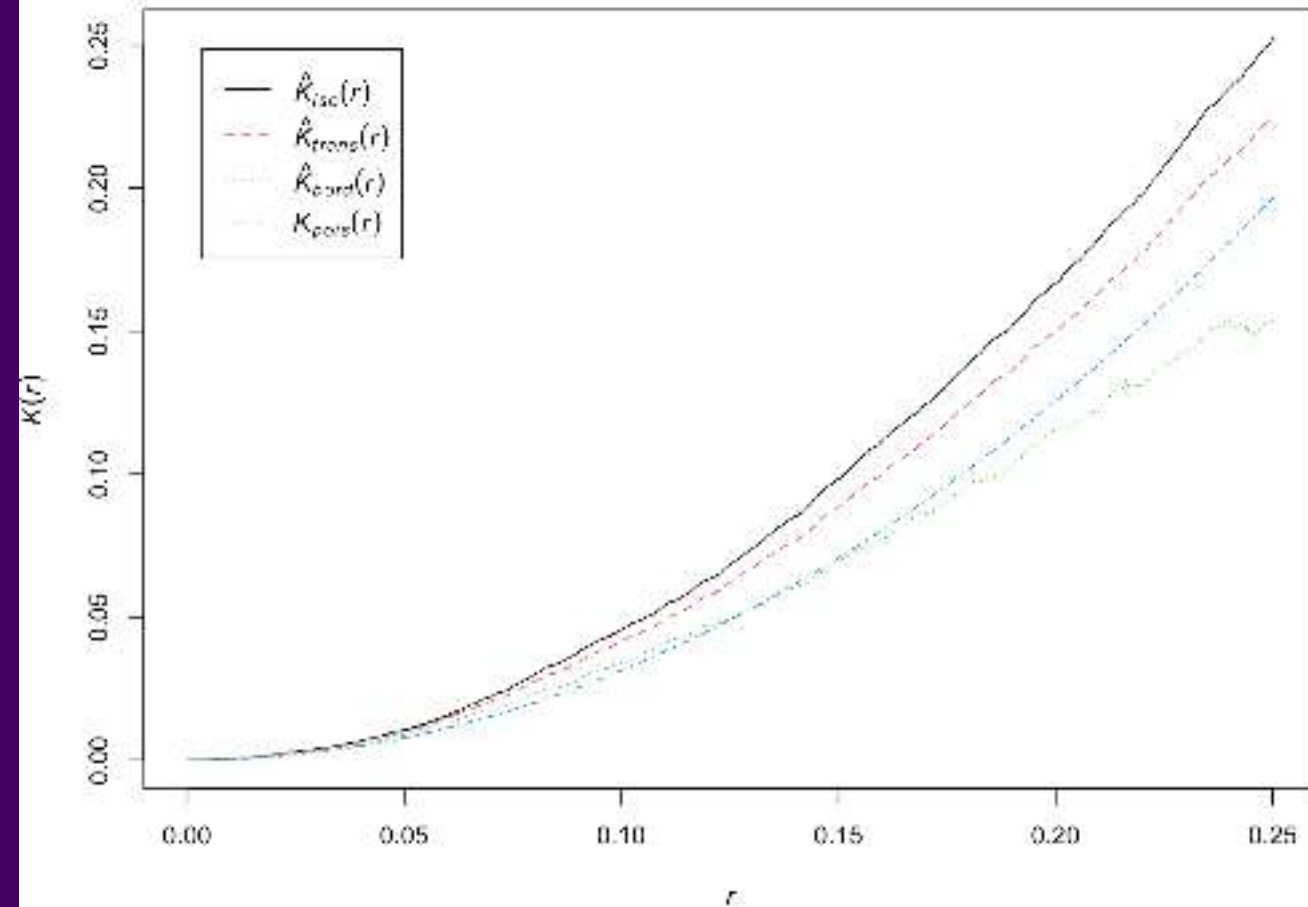


K for IPP

Inhomogeneous Poisson Process (CSR)

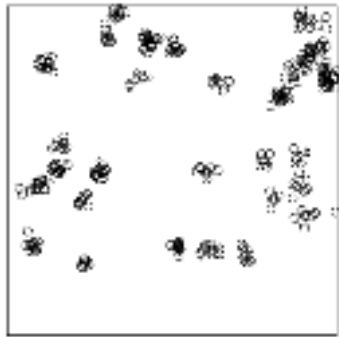


K: Inhomogeneous Poisson

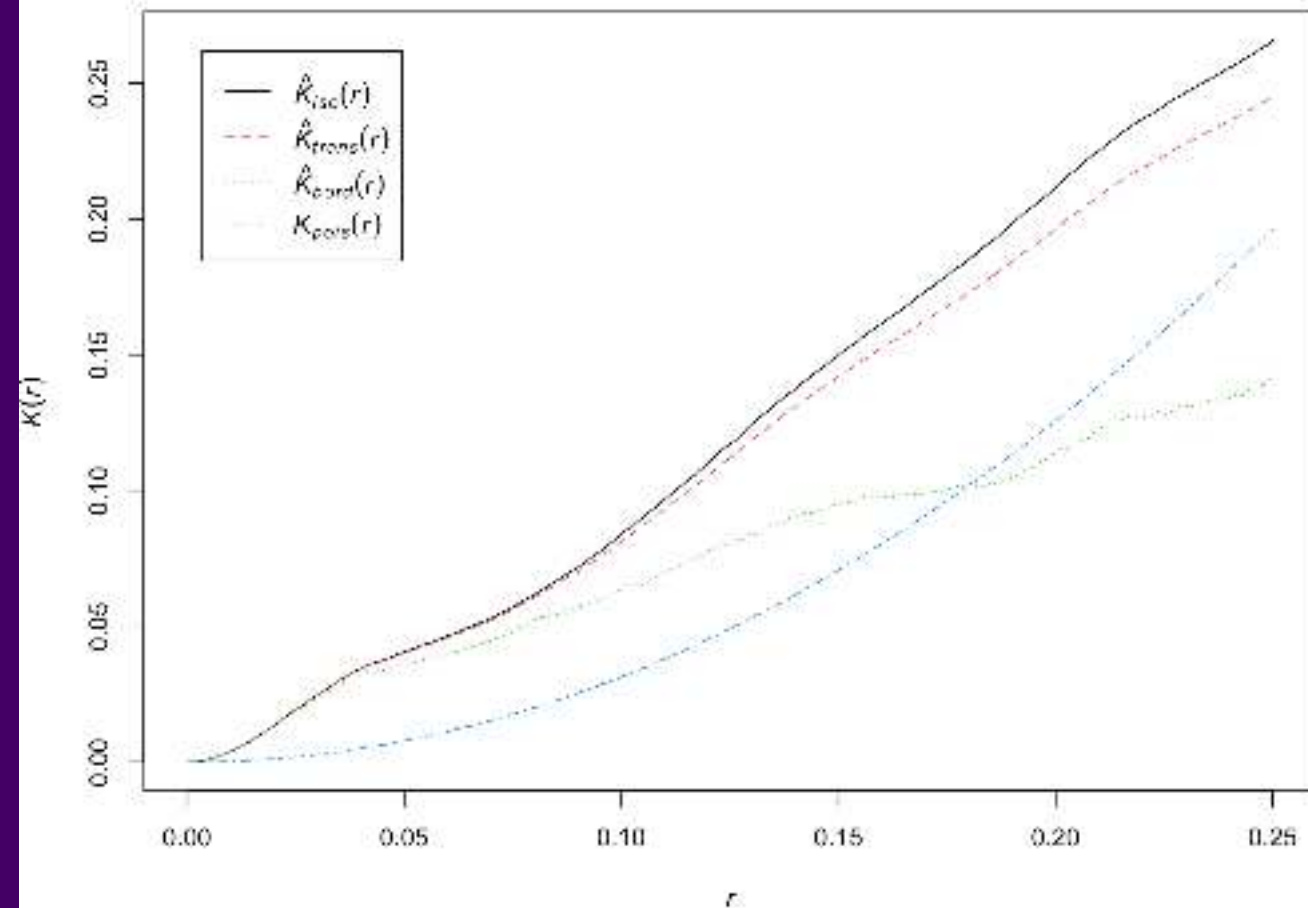


K for Clustered Processes

Clustered Poisson Process (CSR)



K: Clustered Poisson



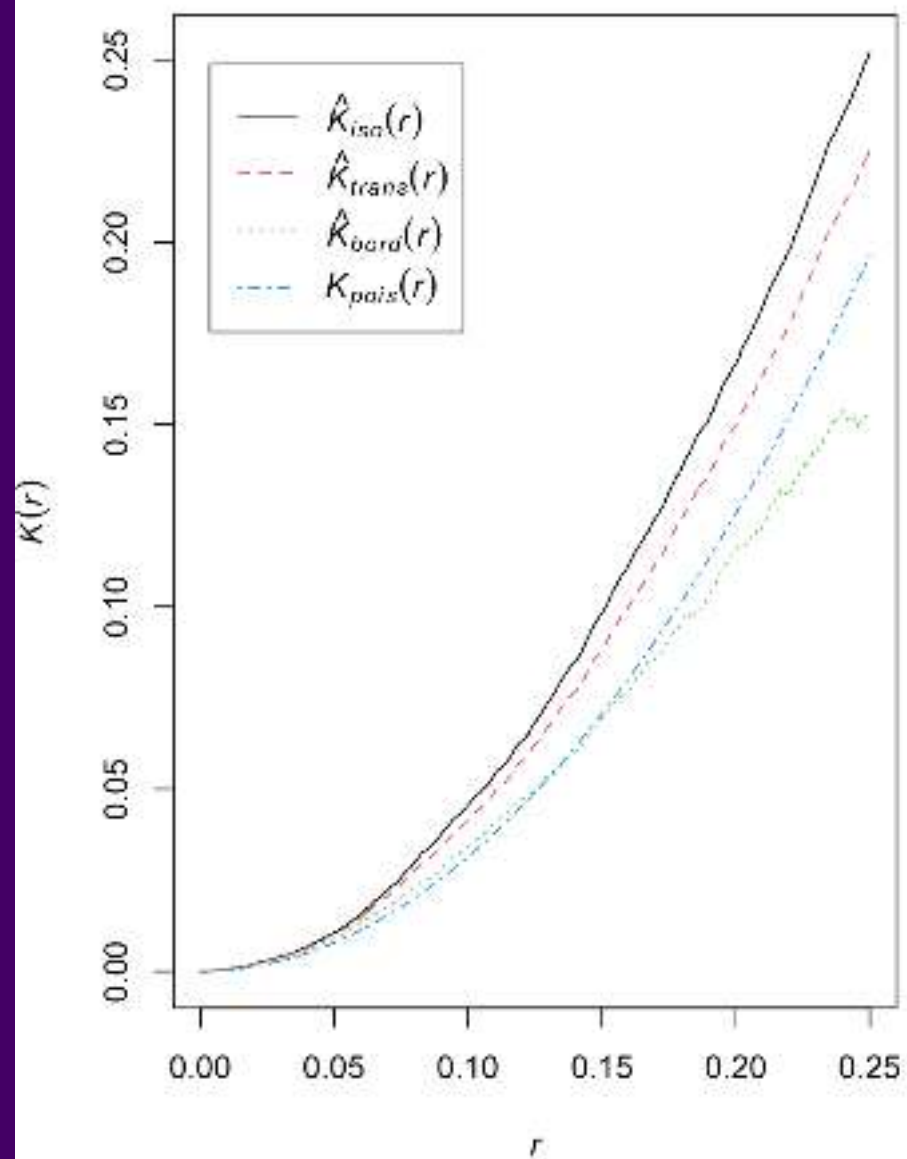
Adjustments for K

- K is sensitive to edge effects (window size)
(**correction** argument to **Kest**)
- K assumes HPP (**Kinhom** function instead of **Kest**)

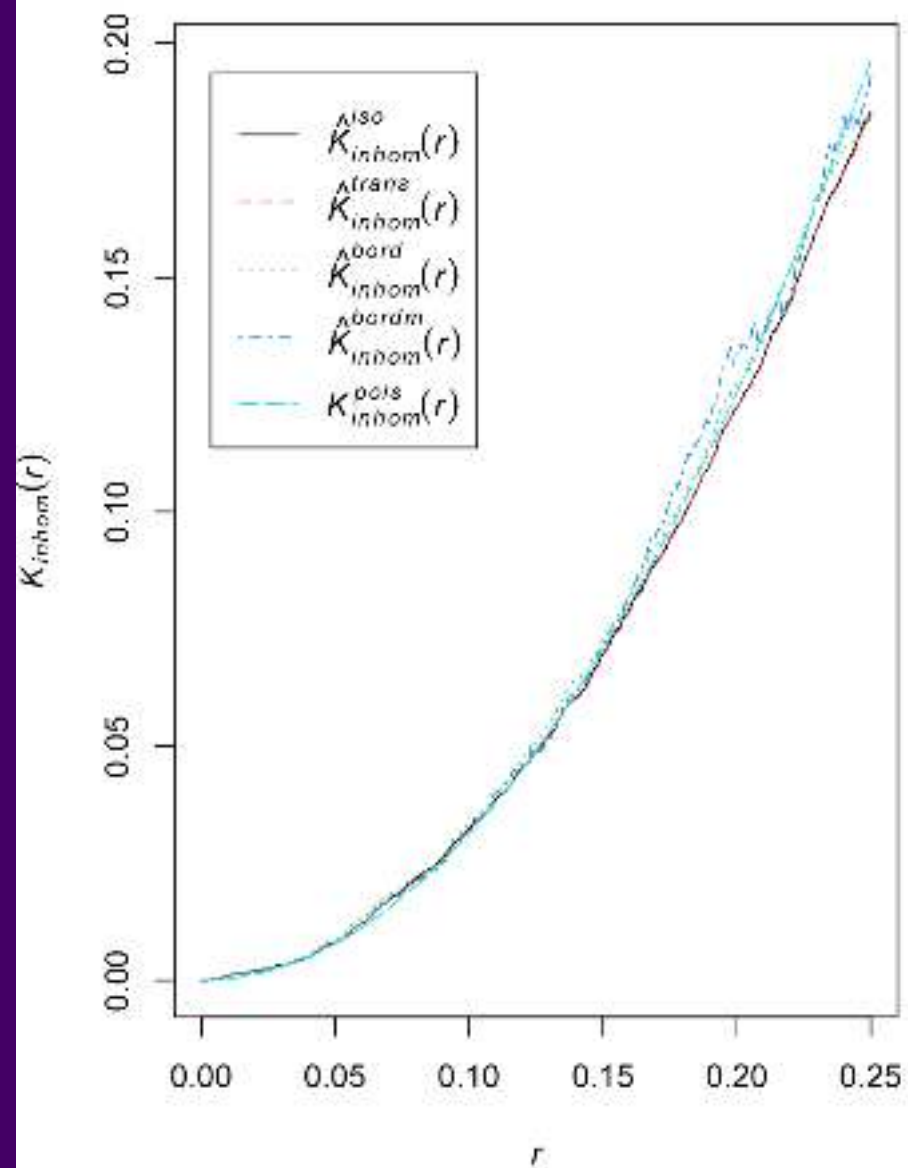
K vs K_{inhom} for IPP



K: Inhomogeneous Poisson

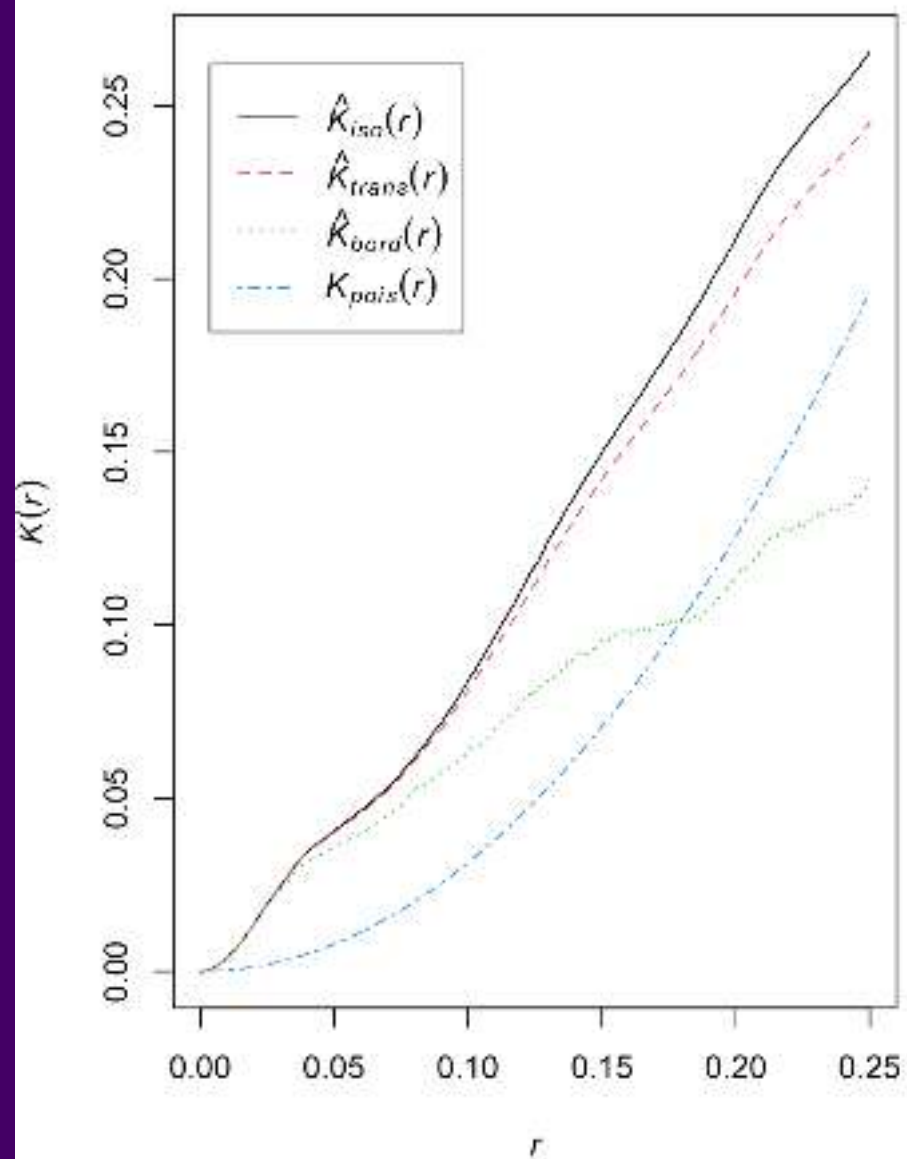


K_{inhom}: Inhomogeneous Poisson

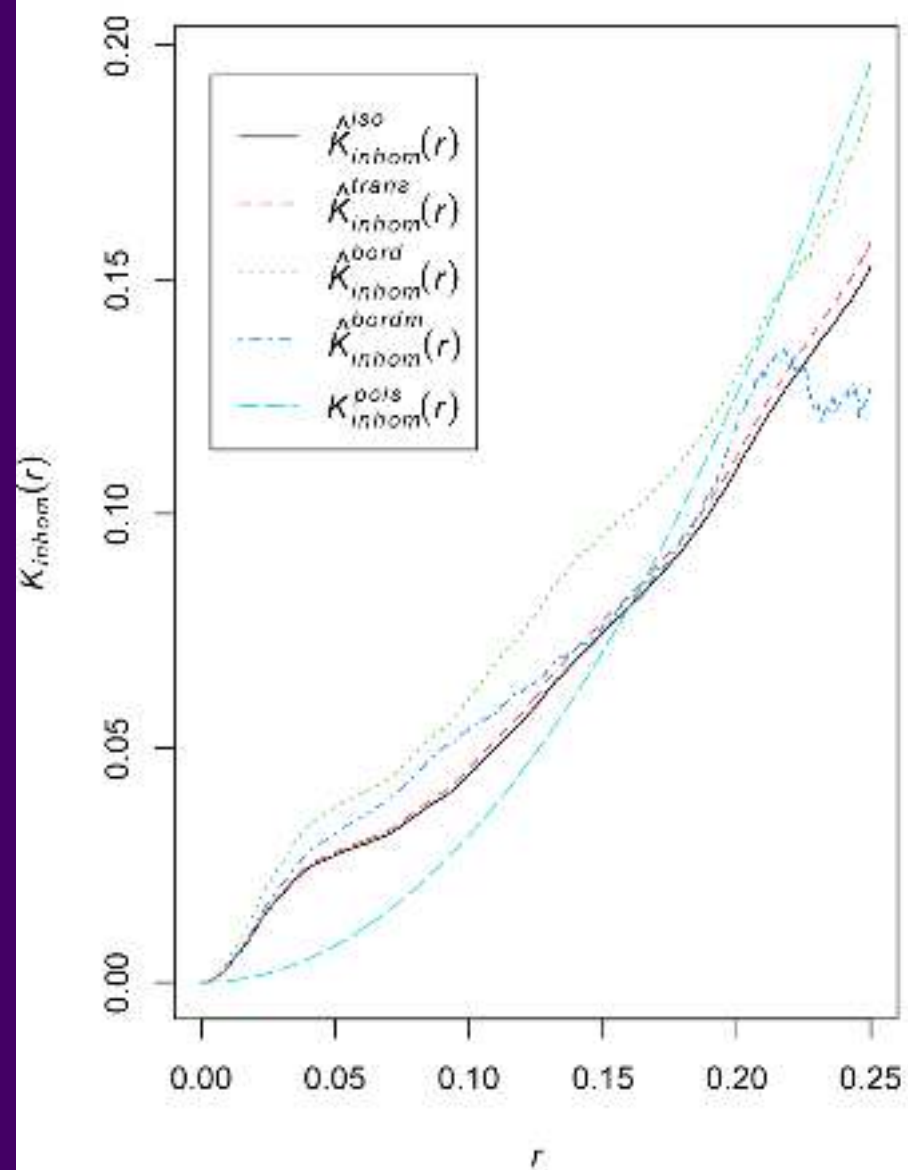


K vs K_{inhom} for Clustered PP

K: Clustered Poisson

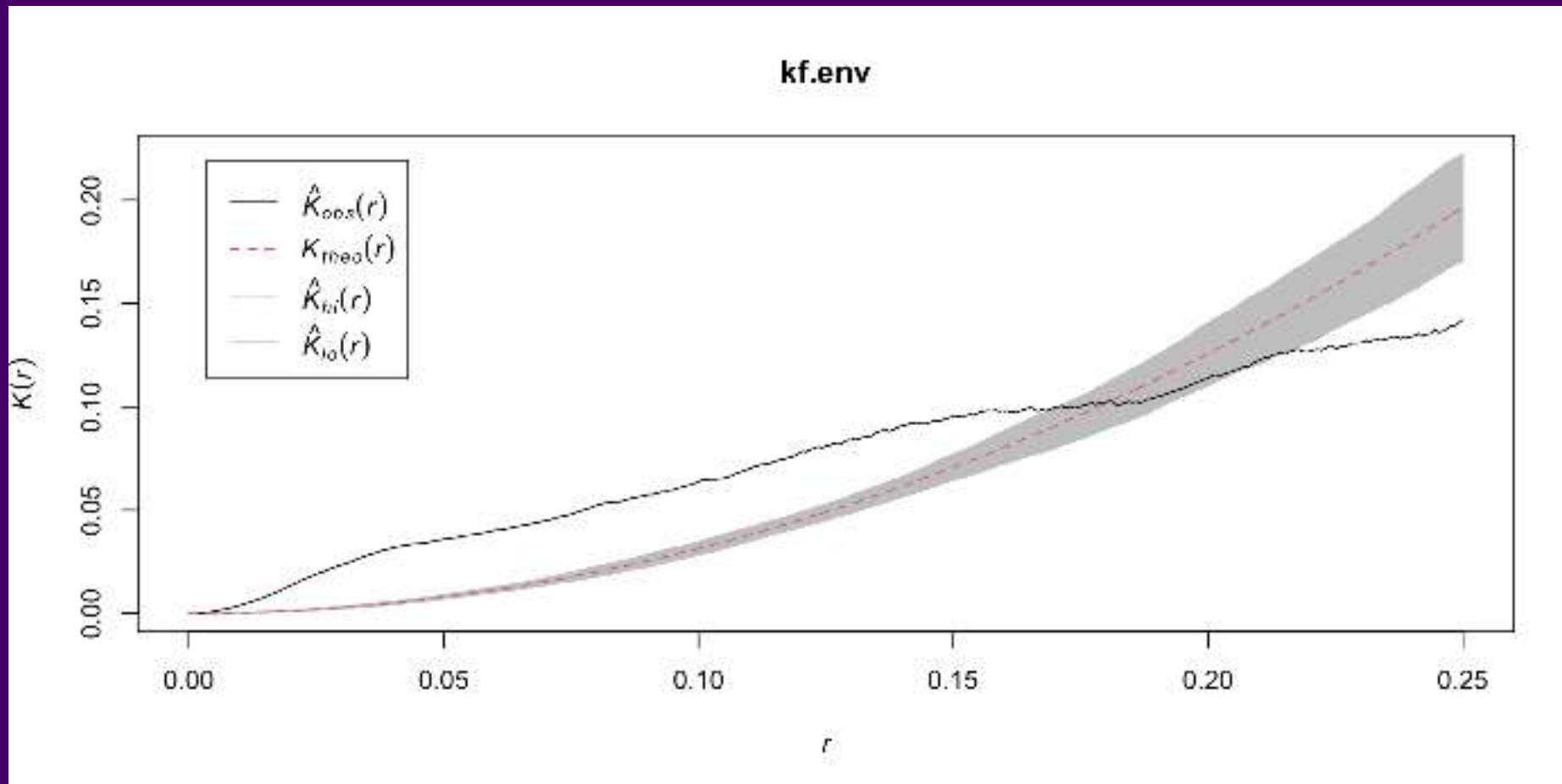


K_{inhom}: Clustered Poisson



Ripley's K Function

- accounting for variation in d



Take-Aways

- Density and Intensity for first-order effects
- Distance for second-order effects
- Linkages for kriging and interpolation

